C-C-305D
October 20, 1971
SUPERSEDING
Int. Fed. Spec. C-C-00305C(Army-GL)
September 30,1970 and
Fed. Spec. C-C-305a
September 28, 1960

FEDERAL SPECIFICATION

CHEESE, PASTEURIZED PROCESS SWISS AND PASTEURIZED PROCESS SWISS AND CHEDDAR, OR AMERICAN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for pasteurized process cheese.

1.2 Classification.

1.2.1 Types. Pasteurized process cheese covered by this specification shall be of the following types as specified (see 6.1):

Type I - Pasteurized process Swiss cheese.

Type II - Pasteurized process Swiss and Cheddar cheese, or Pasteurized process Swiss blended with Cheddar cheese.

Type III - Pasteurized process Swiss and American cheese, or Pasteurized process Swiss blended with American cheese.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 8910

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.

Federal Standard:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items.
MIL-T-43372	-	Temperature of Chilled or Frozen Subsistence.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1162	-	Sanitary Standards for Cheese and Cheese Products Plants.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

U.S. Department of Agriculture

U.S. Standards for Grades of Swiss Cheese

U.S. Standards for Grades of Cheddar Cheese

U.S. Standards for Condition of Food Containers

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to the Information Division, Consumer and Marketing Service, U.S. Department of Agriculture, Washington, D.C. 20250.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

American Public Health Association, Inc.

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th Street, N.W., Washington, D.C. 20036.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

National Association of Dairy Equipment Manufacturers

Recommended Practices for Producing Culinary Steam for Processing Milk and Milk Products

(Application for copies should be addressed to the National Association of Dairy Equipment Manufacturers, 1012 Fourteenth St., N.W., Washington, D.C. 20005.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Cheese. All natural cheese used in the preparation of the process cheese shall have been prepared from milk drawn from cows located in an area designated by the U.S. Department of Agriculture as a modified accredited area or from cows in herds fully accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture or in the process of being accredited. The natural cheese when ready for grinding shall be properly cleaned, free of the following: mold, rind rot, thick dry rind, paraffin, cheese pests, objectionable surface flavor or condition which is unwholesome or which will detract from the quality of the finished product. Cheese made from unheated or unpasteurized milk may be used. All cheese shall be subject to inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.1.1.1 Grades of Swiss cheese and Cheddar cheese. The interpretation of the requirements (descriptive terms) for grades of natural Swiss cheese and natural Cheddar cheese shall be as defined in the U.S. Standards for Grades of Swiss Cheese and U.S. Standards for Grades of Cheddar Cheese, respectively.

3.1.1.2 Swiss cheese, types I, II and III. Natural Swiss cheese used in types I, II, and III may be any grade provided the cheese has a flavor equivalent to grade B or better. Pieces, broken slices and ends of cheese may be used for processing provided they are cut from cheeses meeting the above requirement.

3.1.1.3 Cheddar cheese, types II and III. The natural Cheddar cheese may be of any grade provided the cheese has a flavor equivalent to grade B or better. Cheddar cheese cured in bulk form, commonly termed barrel cheese, may be used, provided it meets the flavor requirements for grade B or better Cheddar cheese. Pieces, broken slices and ends of Cheddar cheese may be used for processing provided they are cut from cheeses meeting the above requirements.

3.1.1.4 Other varieties of cheese, type III only. One or more of the following varieties may be used with Swiss cheese for type III: Cheddar, colby, washed curd, soaked curd, granular or stirred curd. These varieties may be of any grade provided the cheese has a flavor equivalent to grade B or better Cheddar cheese except that tolerances may be made for the characteristic milder flavor attributed to the particular kind of cheese. Pieces, broken slices and ends of cheese may be used for processing provided they are cut from cheeses meeting the above requirements. Varieties of the above cheese in bulk form, commonly termed barrel cheese, may be used provided they meet the flavor requirements for grade B or better Cheddar cheese, except that tolerances may be made for the characteristic milder flavor attributed to the particular kind of bulk cheese.

3.1.2 Cream (including plastic cream). Cream, if used, shall be sweet, pleasing and desirable and free from objectionable flavor. Fresh or frozen cream may be used.

3.1.3 Salt. Salt shall be non-iodized sodium chloride of food grade quality, free from extraneous material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.1.4 Emulsifying agents. Emulsifying agents shall be those permitted by the Definitions and Standards of Identity for Cheeses and Cheese Products under the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.1.5 Acidifying and mold inhibiting ingredients. Acidifying and mold inhibiting ingredients (when used) shall be those, and in amounts permitted by the Definitions and Standards of Identity for Cheeses and Cheese Products under the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.1.6 Water. The water shall be of a safe sanitary quality approved by Federal or State authorities. It may be subjected to suitable chemical or heat treatment and shall not impart undesirable odors or flavors to the finished product.

3.1.7 Culinary steam. Steam in direct contact with the product shall be culinary steam. Culinary steam shall be free from extraneous material and shall be of the quality described in Recommended Practices for Producing Culinary Steam for Processing Milk and Milk Products promulgated by the National Association of Dairy Equipment Manufacturers. The flavor of the steam condensate shall be bland and free from objectionable flavors and odors. The water for the generation of steam may be treated with boiler compounds which conform to the Food Additives, Amendment of Federal Food, Drug, and Cosmetic Act, or a secondary steam generator used in which soft water is converted to steam and no boiler compounds used.

3.2 Preparation.

3.2.1 Type I. Pasteurized process Swiss cheese shall be prepared by comminuting and mixing one or more lots of natural Swiss cheese as specified in 3.1 into a homogeneous mass with the aid of heat, with or without the addition of salt, water, and with the incorporation of not more than 3 percent (based upon the weight of the anhydrous form) of a suitable emulsifying agent. Cream may be added to bring the fat content up to the required percentage but the quantity of the fat derived therefrom shall not exceed 5 percent of the weight of the pasteurized process cheese. Acidifying and mold inhibiting ingredients may be added.

3.2.2 Type II. Pasteurized process Swiss and Cheddar cheese or pasteurized process Swiss blended with Cheddar cheese shall be prepared by comminuting and mixing not more than 75 percent and not less than 51 percent of Swiss cheese and not less than 25 percent of Cheddar cheese into a homogeneous mass as outlined in 3.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.2.3 Type III. Pasteurized process Swiss and American cheese or pasteurized process Swiss blended with American cheese shall be prepared as specified in 3.2.2, except for the type of cheese used.

3.2.4 Pasteurization. The process cheese shall be pasteurized by heating to not less than 158° F. and holding at such temperature for not less than 30 seconds.

3.2.5 Packaging temperature. The process cheese shall be packaged (pouch and carton, can or level C container) at not less than 150° F. After filling and closing, the containers shall be so handled and stacked in such a manner as to permit good ventilation and air circulation. The cheese shall be cooled to a temperature of 100° F. or less within 24 hours from the time the cheese is packaged. Cans may be cooled in water; however, they shall be completely dry prior to packaging into shipping cases.

3.3 Finished product.

3.3.1 All types.

3.3.1.1 Flavor. Type I shall possess a pleasing and desirable flavor and shall be characteristic of process cheese made from good quality Swiss cheese. However, for types II and III the flavor may be somewhat less characteristic of Swiss cheese.

3.3.1.2 Body and texture. The body shall be smooth, medium firm and resilient with practically no gassy pinholes or openings except those caused by entrapped air during filling. The texture shall be close and free from lumps or graininess, slice freely, show no more than slight brittleness or roughness and shall not break when cut. The cheese shall not oil-off when melted.

3.3.1.3 Color. The process cheese shall have a uniform white to creamy white color.

3.3.1.4 Finish and appearance. The process cheese shall be uniform in shape and size with practically straight edges and flat sides, except for canned cheese, slices shall separate readily and in whole slices when removed from the primary container. The product, including canned cheese, shall have dry surfaces, free from mold, cracks, or other surface defects. The liner, wrapper, or pouch shall closely contact the cheese on all surfaces and shall entirely enclose the cheese. The wrapper, pouch, or can shall be completely sealed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

3.3.1.5 Analytical requirements:

	Type I	Types II and III
Moisture (not more than)	44.0 percent	41.0 percent
Fat (in moisture free substance) (not less than)	43.0 percent	47.0 percent
pH (not below)	5.3	5.3
Salt (not more than)	2.5 percent	2.5 percent

Phosphatase (less than) 1 mcg. of phenol per ml. (types I, II and III) of cheese extract, or 0.25 grams of cheese. See 4.5 which provides for optional use of phosphatase test procedure outlined in the Definitions and Standards of Identity for Cheese and Cheese Products.

3.4 Plant qualification. The product shall be prepared, processed and packaged in plants which operate under USDA Regulations Governing Grading and Inspection of Manufactured or Processed Dairy Products and listed by the U.S. Department of Agriculture in the document entitled Dairy Plants Surveyed and Approved for USDA Grading Service and approved for the finished product described in this specification or in plants meeting the requirements of MIL-STD-1162.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder.

3.6 Handling, storage and delivery. Finished product stored for more than one week prior to shipment shall be promptly cooled and held at below 50° F. and during subsequent handling, storage and delivery, the temperature of the product shall not be allowed to rise above 75° F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.1.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Samples of the finished product covered by this specification, and samples of any materials, components, or constituents entering into the preparation of the finished product, may be taken at any time by the Government inspector to determine compliance with the requirements.

4.3 Inspection.

4.3.1 General. Inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, at any suitable point in transit and after delivery to the point of destination.

4.3.2 Condition inspection of containers. When inspection of product for quality and condition is performed by the USDA, the U.S. Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection of containers for the Military, see 4.4.3.)

4.3.3 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification requirements of 3.4.

4.3.4 Factory inspection. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.3.5 Destination inspection. Unless otherwise specified (see 6.2), final inspection shall be made after delivery to the point of destination.

4.4 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.4.1 Component and material inspection. In accordance with 4.1 and 4.1.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified, in this specification or applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.4.1.1 Inspection of dairy products.

4.4.1.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest Office of the Agricultural Research Service of the Animal Health Division. (Location of this office can be ascertained by consulting the publication entitled Directory of Consumer Protection Programs Establishments, Circuits, and Officials.) Nonconformance to either of the above requirements shall be cause for rejection of the lot.

4.4.1.2 Examination of cheese component(s). Unless cheese component(s) is accompanied and identified by a USDA grade certificate in accordance with the pertinent grade B (or better) requirements of 3.1.1.2 through 3.1.1.4, examination shall be made to determine compliance with the flavor requirements of above referenced paragraphs. Each lot of cheese shall be represented by a cheese vat number and date of manufacture. At least one cheese plug drawn from a representative sample of cheese from each vat (lot) to be used in the specified blends shall be examined for flavor. Noncompliance of any lot of cheese with the aforementioned referenced flavor requirements shall be cause for rejection of the involved component lot or finished process cheese made therefrom, as applicable.

4.4.1.3 Water. Certified semiannual laboratory test reports from the state or municipality in which the plant is located or the Federal Government, indicating that the water supply is of a safe sanitary quality, shall be available for review by the Government inspector and may be accepted as evidence of compliance with 3.1.6.

4.4.1.4 Examination of boiler compounds (when applicable). Conformance of boiler compounds to the identity requirements of 3.1.7 shall be ascertained by examination of pertinent labels, invoices or other valid documents. Nonconformance to the above referenced requirements shall be cause for rejection of the involved lot of finished product.

4.4.1.5 Examination of cream (including plastic cream), salt, emulsifying, acidifying and mold inhibiting ingredients. Examination of these components, as applicable, shall be made to determine compliance with the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

requirements of 3.1.2 through 3.1.5 as concerns identity. This determination shall be made by examination of pertinent labels, invoices or other valid documents. In addition, each of the ingredients shall be examined organoleptically, as necessary, to determine conformance to the above referenced requirements and 3.5. The sample for examination shall be approximately 4 ounces of ingredient taken from each of five containers or all containers if five or less form a lot. Nonconformance to identity or condition requirements shall indicate an unacceptable ingredient(s) and use of such shall be cause for rejection of the involved end item.

4.4.1.6 Testing of components. Testing of components described in this specification shall be as shown in table I. Results shall be applicable to the lot average. Nonconformance to test requirements shall be cause for rejection of the lot.

TABLE I. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt paragraph
Cans and lids 1/	1 can and lid	Cans	S-1	Tinplate	Nearest 0.01 lb/ base box	4.5.4 5.1.1.2

1/ Exterior can coating and can label shall be tested in accordance with the applicable subsidiary specification except that tests shall be performed on the same cans submitted for tinplate testing.

4.4.2 In process examination. Unless otherwise specified (see 6.1), examination shall be performed during processing in accordance with 4.4.2.1 and 4.4.2.2.

4.4.2.1 Preparation and controls. The natural cheese components meeting the applicable flavor requirements shall be examined prior to grinding or blending to determine compliance with the cleaning requirements of 3.1.1 and with the preparation requirements of 3.2 and 3.6 as concerns procedure, processing, packaging, handling and storage temperatures and time intervals. Records of all applicable control data shall be maintained and made available

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

for review. Noncompliance with one or more of the above referenced requirements, reflected by actual examination or by examination of records shall be cause for rejection of the involved quantity of finished product.

4.4.2.2 Formulation. Examination shall be made during formulation of the product(s) to determine compliance with the requirements of the applicable product blends and use of permitted components as concerns kind and amounts specified (see 3.2.1 through 3.2.3). Formulation records shall be maintained. Noncompliance with the above referenced requirements, as determined by examination of records or actual examination shall be cause for rejection of the involved quantity of finished product.

4.4.3 Examination of end item. Classification of defects found during examination shall be made in accordance with tables III through V and 4.4.3.1 through 4.4.3.5. The examination criteria shall be as shown in table II. The acceptable quality levels (AQLs) for table III shall be expressed as percent defective and as defects per hundred units for tables IV and V.

TABLE II. End item examination criteria (Military agencies)

Table	Inspection level	Sample unit	Lot size expressed in	AQLs			
				Major A	Major B	Minor	Total
III	S-3	1 primary container	Primary containers	-	-	2.5	-
IV	I	1 primary container	Primary containers	0.25	1.5	-	4.0
V	S-2	1 primary container	Primary containers	1/	1.5	-	6.5

1/ See footnote 1/ of table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

TABLE III. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight more than 1% under specified weight (for units over 5 lbs). (Report weight to the nearest 1/2 ounce.)
202	Net weight more than 2% under specified weight (for units over 1 lb. and up to and including 5 lbs). (Report weight to the nearest 1/4 ounce.)
203	Net weight more than 5% under specified weight (for units 1 lb. or under). (Report weight to the nearest 1/8 ounce.)

1/ Lot shall be rejected if sample data indicate lot average net weight is less than specified net weight.

TABLE IV. Examination of filled pouch and the over-package

Category			Defect
Major A	Major B	Minor	Overpackage
		201	Tear, hole or not completely sealed.
	151		Not clean.
	152		Nomenclature missing, incorrect or illegible
		202	Labeling information other than nomenclature missing, incorrect or illegible.
	153		Missing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

TABLE IV. Examination of filled pouch and the over-package (cont'd)

Category			Defect
Major A	Major B	Minor	
			Pouch
101			Tear, hole or not completely sealed.
	151		Pouch material not adhering very tightly to the product surface.
	152		Not clean.

TABLE V. Examination of product characteristics 1/ 2/

Category			Defect
Major A	Major B	Minor	
101			Presence of foreign material, e.g. dirt, insect(s), insect part(s), wood, paint, glass, metal, etc.
102			Presence of objectionable flavor or odor, e.g. oxidized, bitter, fruity, yeasty, sour, putrid, disinfectant, etc.
			Flavor
	151		Not a pleasing and desirable flavor or not characteristic of process cheese made from good quality Swiss cheese. 3/
			Body and texture
	152		Not smooth, medium firm or resilient.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

TABLE V. Examination of product characteristics 1/ 2/ (cont'd)

Category			Defect
Major A	Major B	Minor	
			Body and texture (cont'd)
	153		Presence of gassy pinholes or openings. 4/
	154		Texture not close and free from lumps or graininess.
		201	Does not slice freely or breaks when cut.
		202	More than slight brittleness or roughness.
	155		Oils off when melted.
			Color
		203	Does not have a uniform white to creamy-white color.
			Finish and appearance
	156		Not uniform in shape and size or shows distortion or lack of conformity to the package shape or has uneven, wet (sticky) surfaces.
	157		Presence of mold, cracks, or other surface defects.

1/ Presence of any Major A defect shall be cause for rejection of the involved lot of finished product.

2/ A standard cheese trier plug of cheese at 70° - 75° F. shall be drawn from each sample unit (see 4.5.2).

3/ For types II and III exception, see 3.3.1.1.

4/ Opening caused by entrapped air during filling shall not be scored as a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.4.3.1 Examination for packaging, liners and interior coating. Examination for packaging, liners and interior coating shall be made in accordance with the applicable examination criteria of PPP-C-29.

4.4.3.2 Examination of can labeling. Examination of can labeling shall be in accordance with examination criteria of MIL-L-1497.

4.4.3.3 Examination of exterior can coating. Examination of cans for coating when required shall be in accordance with the examination criteria of TT-C-495.

4.4.3.4 Examination of shipping containers. Examination shall be made in accordance with the examination criteria contained in the appendix to PPP-B-636 with the following defects added to the table of examination; Major: Marking missing, incomplete or illegible. Minor: (1) Net weight of carton more than 50 lbs. as indicated by marked net weight. (2) Arrangement not as specified. When applicable, containers reinforced with other than pressure-sensitive adhesive, filament reinforced tape shall not be acceptable. For level C pack the aforementioned criteria shall be applicable except that only the defects pertaining to marking and arrangement shall apply.

4.4.3.5 Examination of delivery temperature of product. Examination of delivery temperature of the product shall be made in accordance with the applicable examination criteria (i.e., destination temperature) of MIL-T-43372, except that the temperature limitation of 3.6 shall prevail.

4.4.4 Sampling procedure and acceptance criteria for testing of finished product.

4.4.4.1 Sampling cheese for analytical testing. A 1-pound composite sample shall be formed by taking a proportionate number of cheese plugs from each of the sample units examined for product characteristics. In all cases the sample container shall be filled to capacity to minimize evaporation or condensation until the cheese is tested. Alternatively, proportionate samples of hot cheese may be taken from the processing line to form the one-pound sample unit for each inspection lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.4.4.2 Testing for analytical requirements. The finished product shall be tested for the characteristics specified in 3.3.1.5. Procedures for testing shall be in accordance with 4.5. The results shall be reported as follows: Moisture, fat (in moiture-free substance) and salt -- to the nearest 0.1 percent, pH -- nearest 0.1 unit and phosphatase on pass or fail basis. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

4.5 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Chemical analyses. Chemical analyses, and unless otherwise specified (see 6.1), shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Dairy Products, Section: Cheese.

Test	Method
Moisture	Method I
Fat	Fat
Salt	Total chlorides
pH	Electrometric Method using the Quinhydrone electrode 1/ 2/

1/ Electrometric Method using the Quinhydrone electrode: Grind 2 to 3 grams of cheese with about 0.050 grams of Quinhydrone in a mortar and pestle, and determine the pH with saturated calomel-gold electrodes without any dilution, after adjusting the temperature to 25° C. or correcting to that temperature. Quinhydrone is not considered a diluent.

2/ The Electrometric Method using the glass electrode A.O.A.C. Method may be used optionally but in case of dispute the method given in footnote 1/ above shall be final. A.O.A.C. Chapter 10: Beverages: Malt Beverages and Brewing Materials, Section: Beer, Method: Hydrogen-Ion Concentration; Electrometric Method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

4.5.1.1 Phosphatase. Compliance with the phosphatase requirement shall be determined by the Scharer Rapid Phosphatase Test as outlined in Standard Methods for the Examination of Dairy Products, Chapter: Phosphatase Methods to Determine Pasteurization.

4.5.2 Grading. The processed cheese shall be at room temperature (70° to 75° F.) when examined by the Government inspector for flavor, body, texture and color.

4.5.3 Weighing test. Each shipment or lot of processed cheese may be subject to a weighing test by the procuring agency or duly authorized representative.

4.5.4 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level B or C, as specified (see 6.1).

5.1.1 Level B.

5.1.1.1 Wrapped and cartoned cheese. The product, in units of 8 ounces, 1 pound, 2 pounds, or 5 pounds, as specified (see 6.1), shall be completely sealed in the original pouch into which it was filled and the pouch material shall adhere very tightly to the product surface. Commercial packaging which meets this requirement is acceptable. The pouch shall be over-packaged in a commercial carton.

5.1.1.2 Canned cheese. Seven pounds of the product shall be packaged in an open-top style, round metal can with soldered side-seam and compound-lined, double seamed ends. The cans (approximate size shall be 603 by 700) shall be made throughout from commercial 0.25 pound electrolytic tinplate, and shall be coated inside with an enamel finish furnished by the can manufacturer in accordance with the best commercial practice for the product. The cans shall be fully lined (top, bottom and body) with vegetable parchment or a suitable parchmentized or grease-proof wet-strength paper. The cans shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

5.2 Packing. Not more than 50 pounds of the wrapped and carton cheese or 6 cans of the canned cheese, arranged 3 in length, 2 in width and 1 in depth, shall be packed into a snug-fitting shipping container in accordance with level B or C, as specified (see 6.1).

5.2.1 Level B. The shipping container shall be a fiberboard box, constructed, closed and reinforced in accordance with style RSC or FOL, V3c, V3s or V4s of PPP-B-636. Alternatively, closure of the four flaps on each of the two opposite faces of the style RSC box may be accomplished by making the flaps along the longer edge of the box opening the inner flaps and those along the shorter edge the outer flaps when in the closed position.

5.2.1.1 When specified (see 6.1 and 6.4), the shipping container specified in 5.2.1 shall be reinforced with pressure sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.1.2 When specified (see 6.1 and 6.5), the shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC or FOL, Wfc or W5s of PPP-B-636. Alternatively, closure of the four flaps on each of the two opposite faces of the style RSC box may be accomplished by making the flaps along the longer edge of the box opening the inner flaps and those along the shorter edge the outer flaps when in the closed position.

5.2.2 Level C. The shipping container shall be in accordance with the Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Labeling and marking.

5.3.1 Civil agencies.

5.3.1.1 Unit containers. Unless otherwise specified (see 6.1), any labeling of unit containers that complies with the Federal Food, Drug and Cosmetic Act, and Regulations Promulgated Thereunder, is acceptable.

5.3.1.2 Shipping containers. Unless otherwise specified (see 6.1), shipping containers shall be marked in accordance with FED-STD-123. Further, the container shall be marked with the quantity and size of packages in the container, and have the following precautionary marking stenciled or printed on the top of the shipping containers in letters between 1-1/2 and 2 inches high, all letters being of the same height:

PERISHABLE - KEEP REFRIGERATED

In addition, such other marking requirements as specified by individual agencies shall be included.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

5.3.2 Military agencies.

5.3.2.1 For cheese sealed in a protective pouch and carton. Commercial labeling will be acceptable.

5.3.2.2 Cans. The cans of the product shall be labeled in accordance with MIL-L-1497.

5.3.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the shipping container for wrapped and cartoned cheese shall have the following precautionary marking stenciled or printed on the top of the shipping container in letters between 1-1/2 and 2 inches high, all letters being of the same height:

KEEP REFRIGERATED
(40° TO 45° F.)
DO NOT FREEZE

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Type of cheese required (see 1.2).
(b) When chemical analysis is not required (see 4.5.1).
(c) Size of unit required for commercially packed product (see 5.1.1.1).
(d) Selection of applicable level of packaging and packing (see 5.1 and 5.2).
(e) When packing in accordance with 5.2.1.1 or 5.2.1.2 is required.
(f) When a USDA inspection or grade certificate is required (see 4.4.1.2).
(g) When in process examination is not required by Civil agencies (see 4.4.2).
(h) When labeling other than specified in 5.3.1.1 is required.
(i) When shipping containers are to be marked other than in accordance with FED-STD-123 (see 5.3.1.2)

6.2 Destination inspection (for civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for identity, condition and quantity only.

6.3 Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15.

6.4 Packing specified in 5.2.1.1 is intended for transfer at sea operations and specific overseas operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

6.5 Packing specified in 5.2.1.2 may be used when experience or knowledge of the destination and mode of shipment would indicate that a lower grade of board can be used.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST

VA
AGR
HEW
DC
INT
GSA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 20 cents each.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1974-603-120/71

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

U.S.MAIL

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION C-C-305; Cheese, Pasteurized Process Swiss and Pasteurized Process Swiss and Cheddar, or American

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:05Z
Check the source to verify that this is the current version before use.